



The bone house

This reconstruction of a shelter built from mammoth bones is based on the remains of a dwelling that was found by archaeologists in modern-day Ukraine. It demonstrates the ingenuity and adaptability of early humans to local conditions and resources.

Europe. These Cro-Magnon people—named after a rock shelter near Les Eyzies in southwestern France—were opportunists. They relied for their subsistence on a range of plant foods and fish, taking advantage of salmon runs, for example, when the rapidly changing climate of the late Ice Age allowed. Their success came not only from their superior mental abilities, but also from their ingenious multi-purpose flint tools, which worked almost like a modern Swiss Army knife. They used carefully shaped flint nodules to produce standardized, parallel-sided blanks, which they then turned into points, scrapers, and other tools. One of these artifacts—a chisel—allowed them to cut grooves in reindeer antlers, thereby “unlocking” a new technology for manufacturing harpoon heads, spear points, and other hunting weapons. Barbed, antler-tipped spears were especially effective on reindeer and other game. The Cro-Magnons produced other revolutionary items, including the spear thrower—a hooked stick that vastly increased the distance a spear could be thrown. They successfully used this new technology to hunt a

wide range of Ice Age animals, including bison, mammoth, and woolly rhinoceros. The eyed needle was another remarkable invention (see below). These people were also skilled artists and developed a distinctive visual tradition, which amplified their elaborate rituals and beliefs. One of the most famous examples of their art comes from the cave paintings of Lascaux in southwestern France, which are on a huge scale, and renowned for the skill of the artists who created them

» Sometime after about 50,000 years ago, when glacial conditions in the north had improved and the climate was more temperate, modern humans moved into Europe and Asia. Tiny numbers of people were involved—in the hundreds—but by 45,000 years ago they were well established in the eastern European plains and in the Don Valley, now in Ukraine, and were moving rapidly across Central and Western Europe.

The Neanderthal controversy

Homo sapiens had settled alongside Neanderthal bands that had already been in Europe for about 200,000

years. DNA research on Neanderthal bones suggests that the newcomers did not interbreed with them, as had previously been believed. One theory is that Europe’s indigenous inhabitants died out because they lacked the adaptability, mental abilities, and technology of modern humans. They survived in some parts of southeastern Europe until as late as 24,000 years ago before becoming extinct.

A thriving European culture

From about 40,000 to 10,000 years ago, a remarkable array of sophisticated, cold-adapted hunter-gatherer societies flourished in Central and Western

INVENTION

THE NEEDLE

The eyed needle was a groundbreaking invention. As early as 30,000 years ago, late Ice Age people in Europe and Asia made needles from polished bone and ivory slivers, perforated with sharp-pointed flints. They sewed tailored, layered garments that enabled them to work outside in freezing temperatures. It is believed that, like modern Inuits, they used cured and softened animal pelts, sewing the seams with fine thread made of animal and plant fiber. Without tailored clothing *Homo sapiens* would never have settled the Eurasian steppes or colonized the Americas.



The longest journey

The first *Homo sapiens* left Africa to colonize the planet about 60,000 years ago. By the end of the Ice Age 10,000 years ago, they had reached all the continents except Antarctica, adapting to different conditions wherever they went.

